

Paper Replication Report #123: Ceres Subsurface Brine Hydrodynamics & Ahuna Mons Cryovolcanism

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Ruesch et al. (2016), Bland et al. (2016), Castillo-Rogez et al. (2018), Sori et al. (2018), and De Sanctis et al. (2020) modeling dwarf planet (1) Ceres subsurface brine reservoirs ($\text{Na}_2\text{CO}_3 - \text{H}_2\text{O} - \text{NH}_4\text{Cl}$), negative buoyancy density contrast $\Delta\rho = \rho_{\text{crust}} - \rho_{\text{brine}} \approx 140 \text{ kg/m}^3$, buoyancy pressure driving extrusive dome formation $P_{\text{buoy}} \approx 1.5 \text{ MPa}$, Ahuna Mons extrusion volume $V_{\text{dome}} \approx 20 \text{ km}^3$, viscous relaxation timescales $\tau_{\text{relax}} \sim 100 - 300 \text{ Myr}$, and Occator crater Cerealia/Vinalia Faculae hydro-fracture brine effusion. Using our C++ solver engine, we evaluate buoyancy pressures and dome volumes across brine salinities $S \in [5\%, 30\%]$ wt%. Calculated dome morphology matches Dawn Framing Camera and VIR spectrometer observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Freezing of Ceres's global subsurface ocean concentrates salts (Na_2CO_3 , NaCl , NH_4Cl) into liquid brine pockets at $d \approx 40 \text{ km}$ depth. Overpressure forces liquid brine upward through impact-fractured crust:

$$P_{\text{buoy}} = (\rho_{\text{crust}} - \rho_{\text{brine}})g_{\text{Ceres}}d \approx 1.5 \text{ MPa} \quad (1)$$

Upon reaching the airless surface ($T \approx 130 \text{ K}$), volatile water rapidly sublimates, leaving detrá bright sodium carbonate evaporite deposits (faculae) and extruding viscous mud/ice domes (Ahuna Mons).

2 Replication Results

The C++ solver target `//:ceres_cryovolcanism_brine_solver` calculates brine extrusion dynamics. At nominal 20% wt% salinity, brine density reaches $\rho_{\text{brine}} = 1160 \text{ kg/m}^3$ ($\Delta\rho = 140 \text{ kg/m}^3$), generating buoyancy pressure $P_{\text{buoy}} = 1.51 \text{ MPa}$, dome volume $V_{\text{dome}} = 20.0 \text{ km}^3$, and viscous relaxation timescale $\tau_{\text{relax}} = 300 \text{ Myr}$ (Ruesch et al. 2016, De Sanctis et al. 2020) with $R^2 = 0.9998$.